

Applied Analytics Project
March Madness Code Output Review
Geovanni Jones, Joshua Previte and Casey Barrasso

SETUP, DATA INGESTION, AND DATA CLEANING

The setup including importing libraries and defining project paths does not produce any unnecessary outputs.

For data ingestion, the shape of all three data files is clearly printed, making it easy to understand the scope of each unique file. Printing the head of the dev_seasons data makes it easy to see many of its features. While the shape of the other two files is helpful, it could be beneficial to include more about each of these, like printing the head of each, to help viewers understand their features.

During the data cleaning phase, the print commands adequately summarize all of the steps of cleaning.

FEATURE ENGINEERING, EDA, AND PREPROCESSING VALIDATION

In the feature engineering phase, code output matches the description of the features that are being created. Much like in the data cleaning phase, the simple print commands allow the viewer to clearly follow how the code is altering the data.

The EDA section is brief, but also includes the plots that are most relevant to finding a solution to the problem. Since both plots are both helping the viewer understand complex relationships between features, something could be done to make them easier to understand. In the first plot, for example, the feature names could be simplified. In both plots, the x-axis labels could be re-written to be more intuitive.

The output in the preprocessing validation section uses print commands well to confirm that the steps taken during data cleaning were successful.

MODELING

At each step of the modeling phase, each model's performance is described well with outputs containing tables of performance metrics. These charts clearly show the differences in performance between train, validation, and test data, as well as between models. The feature importance table in the model explainability section is particularly useful. The only thing that might be helpful to add to the output are confusion matrices to help the viewer visualize the model's performance. Also, since ROC-AUC is used as an evaluation tool, a graph that shows the ROC curve could also be used.

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BIAS DETECTION AND DEPLOYMENT

The bias mitigation segment output provides a clear and concise visual of how rebalancing the data changes the way the model performs on each of the three groups.

The output in the deployment section is also clear and concise and supplements the initial summary of how the model's performance will be monitored during deployment. The final piece of output is a strong way of presenting to viewers what this model would look like to users.